Ogbonna Arinze

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SUMMARY

Senior Frontend Developer with 8+ years of experience building complex, performance-critical web applications using React, TypeScript, and Redux in large-scale enterprise environments. Proven track record of modernizing legacy frontend systems, architecting extensible component libraries, and optimizing real-time UI for large datasets. Deeply passionate about CI/CD tooling and developer experience, with hands-on expertise in frontend architecture decisions, automated testing strategies, and close collaboration with UI/UX designers to deliver polished, high-performance interfaces.

CORE COMPETENCIES

- **React & TypeScript:** React 18/19, component lifecycle, hooks, memoization, concurrent rendering, virtual DOM internals, and advanced TypeScript patterns for type-safe UI development.
- **State Management & Data Fetching:** Redux (Toolkit, RTK Query), React Query, Zustand, GraphQL, REST API integration, and efficient cache invalidation strategies.
- **Performance Engineering:** Rendering optimization, virtualization for large datasets, code splitting, lazy loading, bundle analysis, React Profiler, Chrome DevTools, Lighthouse, and Core Web Vitals.
- **Testing & Quality:** Jest, Cypress, Playwright, end-to-end and integration testing, visual regression testing, CI-driven QA, and Allure test reporting.
- **Frontend Architecture:** Extensible component systems, design systems, plugin-based architectures, micro-frontends, monorepos, and SSR/SSG strategies.
- **JavaScript & Browser Internals:** Deep understanding of JavaScript engine behavior, event loop, browser and DOM APIs, HTML5, CSS3, and cross-browser compatibility.
- **CI/CD & DevOps:** CI/CD pipeline design and automation, GitHub Actions, Docker, AWS (CloudFront/S3), Vercel, and build pipeline optimization with Webpack/Vite/Rollup.
- **Design Collaboration & UX:** Figma-to-code workflows, design system implementation, accessibility (WCAG/ARIA), and close partnership with UI/UX designers.
- **Leadership & Collaboration:** Technical direction, architecture reviews, mentorship, Agile/Scrum, TDD/BDD, code reviews, and cross-functional team collaboration.

WORK EXPERIENCE

Roche Genentech — San Francisco, CA (Remote)

Frontend Software Engineer (React/TypeScript) — *May 2023 - Present*

- Architected a **reusable, extensible design system** in **React** and **Radix UI** with **TypeScript**, establishing a scalable component library with plugin-style composability that improved UI consistency, accessibility, and developer velocity across multiple product teams.
- Developed new features and **optimized real-time UI performance** in a **React** and **TypeScript** SPA handling large clinical trial datasets, implementing **virtualization, lazy loading, code splitting**, and dynamic imports to reduce initial load times by over 40%.
- Diagnosed and resolved complex **rendering bottlenecks** at both component and architectural levels using **React Profiler**, **Chrome DevTools Performance** tab, and **Lighthouse audits**, achieving measurably smoother interactions across data-intensive views.
- Scaled **state management** in a large application by integrating **Redux Toolkit** for complex client state and **React Query** for server-side cache, eliminating redundant re-renders and establishing consistent, maintainable data flow patterns that accelerated developer onboarding.
- Participated in **frontend architecture decisions**, leading the evaluation and adoption of new patterns for component composition, state management strategies, and API integration layers, helping the system evolve incrementally without disruption.
- Engineered **JWT-based authentication** and **role-based access control** in a **Next.js** frontend with **TypeScript** and **React Query**, implementing secure HTTP-only cookie token handling and permission-driven UI rendering in collaboration with backend teams.

- Modernized a **legacy Vue.js web application** into a **React-based Chrome extension**, refactoring legacy code while preserving business logic, optimizing workflows for clinical site integrations, and enhancing adoption through iterative improvements.
- Increased release reliability by integrating **Playwright end-to-end tests** and **Jest unit tests** into the **CI/CD pipeline**, automating test reporting with **Allure Reports** hosted on GitHub Pages, and configuring real-time Slack notifications for build quality visibility.
- Collaborated closely with **UI/UX designers** to translate Figma prototypes into pixel-perfect, accessible React components, iterating on design feedback loops and ensuring consistent implementation of the design system across all product surfaces.
- Improved **CDN asset delivery** and caching strategies using **AWS CloudFront**, optimizing static resource loading and reducing time-to-interactive for users on slower network connections, measured through **Core Web Vitals** and **Datadog RUM** monitoring.

DESCASIO LTD Lagos, Nigeria
Staff Frontend Engineer *May 2021 - Apr 2022*

- Owned the core **frontend architecture** for a data-heavy **React** dashboard serving enterprise clients, implementing **virtualization** and efficient **Redux** state management to handle tens of thousands of rows with smooth scrolling and responsive interactions.
- Optimized frontend performance by profiling and resolving render bottlenecks using **React Profiler** and **Lighthouse**, reducing bundle size by **30%-40%** through **webpack-bundle-analyzer** and **rollup-plugin-visualizer**, achieving a **35% faster load time** across devices.
- Designed and implemented an **extensible component architecture** with clearly defined interfaces and composition patterns, enabling teams to add new dashboard widgets and data views without modifying the core framework.
- Built **real-time data update** capabilities using **WebSockets** and efficient DOM reconciliation strategies, ensuring the dashboard reflected live data changes without full page reloads or unnecessary re-renders.
- Developed **comprehensive testing strategies** including unit tests with **Jest**, integration tests, and end-to-end tests with **Cypress**, achieving high test coverage that significantly reduced production regressions.
- Implemented **algorithmic optimizations** for client-side data filtering, sorting, and search across large datasets, applying efficient data structures to maintain sub-second response times in the UI layer.
- Established engineering standards for **code quality, testing**, and deployment across the team using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing lint errors by 90% and accelerating **CI/CD** cycle times.
- Built custom **Webpack and Rollup** build configurations optimized for the dashboard's module structure, implementing tree shaking, scope hoisting, and chunk splitting strategies that minimized bundle sizes for production deployments.
- Partnered with **UI/UX designers** throughout the development cycle, participating in design reviews, providing technical feasibility feedback, and ensuring seamless delivery of complex data visualization components.
- Mentored junior engineers and guided **frontend best practices**, conducting code reviews focused on **React** performance patterns, **TypeScript** type safety, and **testing discipline**, fostering a culture of high-quality, maintainable code.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*